Deep Learning-based Channel Estimation for Massive MIMO-OTFS Communication Systems

Machine Learning for Communication (EEL7480) — Course Project Report

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***Abstract*—This paper presents a reproduction and critical analysis of the deep learning (DL)-based channel estimation framework proposed by Payami and Blostein [1] for high-mobility massive MIMO-OTFS wireless networks. The original framework comprises three convolutional neural networks (CNNs): PositionNet for support detection, PositionNet$_r$ for residual refinement, and AmplitudeNet for gain estimation. In this work, we reproduce the first stage, PositionNet, using TensorFlow/Keras with a fully Python-based data generation pipeline using the Extended Vehicular A (EVA) channel model, replacing the MATLAB-based dataset generation used by the original authors. Beyond reproduction, we contribute two original extensions: (i) a *hardware-aware architectural correction* that identifies and resolves a parameter-count inconsistency between the paper's stated 64-filter design and its claimed 3 800-parameter footprint, adjusting the filter count to 51 to strictly satisfy this budget; and (ii) a *multi-metric diagnostic evaluation framework* that exposes the precision-recall trade-off concealed by the single accuracy metric reported in the original work. Our reproduced PositionNet achieves 94.4% precision, closely matching the original result, while our expanded evaluation reveals a recall of 54.4%, an F1-score of 0.690, and a cosine similarity of 0.736 on the validation set. These findings provide a rigorous diagnostic justification for the multi-stage architecture of the full pipeline.**

***Index Terms*—OTFS, massive MIMO, channel estimation, deep learning, PositionNet, convolutional neural network, delay-Doppler domain, sparse signal recovery, 6G.**

I. Introduction

Providing reliable communications in high-mobility applications— vehicular communications, high-speed rail, and low-Earth-orbit satellite links—is an important goal of beyond-5G and 6G wireless cellular networks [2], [3]. Currently, orthogonal frequency division multiplexing (OFDM) is the dominant modulation technique for 5G networks. However, in high-mobility applications, OFDM experiences considerable inter-carrier interference (ICI) due to Doppler shifts of time-varying channels, also known as doubly-dispersive channels [4].

Orthogonal Time Frequency Space (OTFS) is a recently introduced modulation technique that modulates information-bearing symbols on the delay-Doppler (DD) plane [5], [6]. By employing the inverse symplectic finite Fourier transform (ISFFT), OTFS transforms frequency-selective time-varying fading channels into quasi-time-invariant channels, making it highly suitable for high-mobility scenarios. An important advantage of OTFS is its ability to compactly represent the channel impulse response of doubly-dispersive channels, and this channel sparsity is the key property that enables efficient estimation [7], [8].

To realize OTFS's benefits, accurate channel state information (CSI) is required at the base station. In the massive MIMO regime, the joint estimation problem becomes high-dimensional. Classical compressive sensing methods such as Orthogonal Matching Pursuit (OMP) [9] and 3D-SOMP [10] achieve accurate CSI but at prohibitive computational cost, with each iteration requiring the Moore-Penrose pseudo-inverse of a large matrix.

Payami and Blostein [1] proposed a three-CNN cascade to solve this problem end-to-end: (1) PositionNet for support detection, (2) PositionNet$_r$ for residual refinement, and (3) AmplitudeNet for gain estimation. This approach achieves an 80% reduction in computation time compared to 3D-SOMP, with BER performance close to perfect-CSI MMSE estimation.

Our contributions in this project are:

- Reproduction of PositionNet (Stage 1) in TensorFlow, with a fully Python-based data generation pipeline replacing the original MATLAB-based dataset, achieving 94.4% precision consistent with the original results.
- Identification and correction of an internal inconsistency between the paper's stated 64 feature maps and claimed 3 800-parameter budget, correcting the filter count to 51.
- A multi-metric evaluation framework (precision, recall, F1, Top-K accuracy, cosine similarity) that reveals a 94.4%/54.4% precision-recall imbalance, providing a quantitative justification for the multi-stage pipeline design that is absent from the original paper.

II. OTFS Preliminaries and System Model

A. OTFS Modulation

Let N and M denote the number of symbols along the Doppler and delay axes, respectively. The 2D DD-domain grid contains NM grid points with Doppler spacing $\Delta\nu = \nu_r/N$ and delay spacing $\Delta\tau = \tau_r/M$.

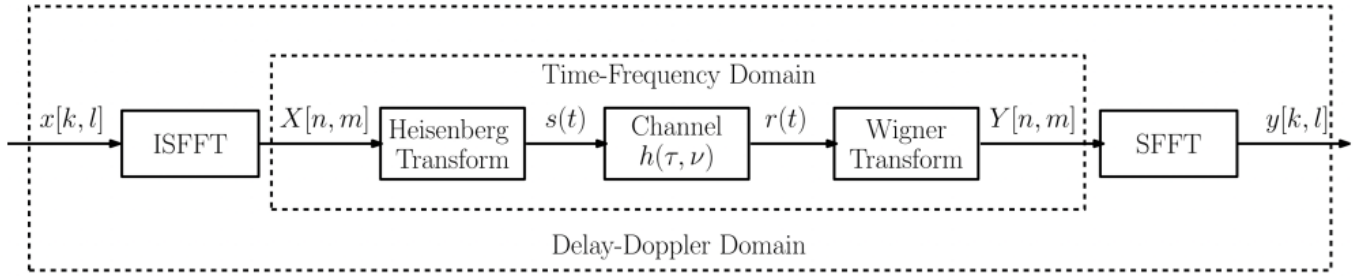

Fig. 1. OTFS modulation and demodulation chain in the delay-Doppler and time-frequency domains [1].

The MN data symbols $x[l,k]$ are first placed in the DD domain. The ISFFT transforms them to the time-frequency domain:

$$X[m,n] = \frac{1}{\sqrt{MN}} \sum_{l=0}^{M-1} \sum_{k=0}^{N-1} x[l,k]\, e^{j2\pi\left(\frac{nk}{N} - \frac{ml}{M}\right)}. \quad (1)$$

The Heisenberg transform then converts $X[m,n]$ to a time-domain signal which is transmitted through the wireless channel. At the receiver, the Wigner transform followed by the SFFT recovers the DD-domain received signal $y[l,k]$ [12].

B. Massive MIMO-OTFS System Model

We consider a massive MIMO-OTFS uplink with N_t transmit antennas at the user and N_r receive antennas at the base station. The delay-Doppler domain channel, following the EVA model, is expressed as [11]:

$$\mathbf{H}(\tau,\nu) = \sum_{q=1}^{N_q} \mathbf{H}_q\, \delta(\tau - \tau_q)\, \delta(\nu - \nu_q), \quad (2)$$

where N_q is the number of multipath scatterers, and $\mathbf{H}_q \in \mathbb{C}^{N_r \times N_t}$ contains the complex gains. The channel matrix in the angular domain is $\mathbf{H}_q = \sqrt{\frac{N_t N_r}{N_q}}\, h_q\, \mathbf{a}_t(\phi_q)\mathbf{a}_r(\theta_q)^H$, where ϕ_q, θ_q are the angle of departure (AoD) and angle of arrival (AoA), and $\mathbf{a}_t(\cdot)$, $\mathbf{a}_r(\cdot)$ are the ULA steering vectors.

The end-to-end system model for single-antenna users is formulated as the following sparse signal recovery problem [10]:

$$\mathbf{y}_{\mathrm{DD}} = \mathbf{\Phi}\, \mathbf{h}_{\mathrm{ADD}} + \mathbf{z}_{\mathrm{DD}}, \quad (3)$$

where $\mathbf{y}_{\mathrm{DD}} \in \mathbb{C}^{M_\tau N_\nu \times 1}$ is the received pilot signal, M_τ and N_ν are the numbers of pilots along the delay and Doppler dimensions, $\mathbf{\Phi} \in \mathbb{C}^{M_\tau N_\nu \times M_\tau N_\nu N_t}$ is the sensing matrix formed by complex Gaussian pilot sequences, $\mathbf{h}_{\mathrm{ADD}} \in \mathbb{C}^{M_\tau N_\nu N_t \times 1}$ is the sparse angle-delay-Doppler channel vector, and $\mathbf{z}_{\mathrm{DD}}$ is Gaussian noise.

C. Simulation Parameters

The simulation parameters follow Section IV of [1] and are listed in Table I.

TABLE I
SIMULATION PARAMETERS [1]

Parameter	Value
Carrier frequency f_c	2.15 GHz
Base OFDM FFT size M	1024
Delay bins M, Doppler bins N	600, 10
Subcarrier spacing Δf	15 kHz
Max. user velocity $v_{\max}$	360 km/h
Transmit antennas N_t	16
Pilot overhead	25%
Channel model	3GPP EVA (6 taps)
Training pilots	Complex Gaussian
Dataset size	10^5 samples
Train / Validation split	70% / 30%
Training SNR	10 dB

III. PROPOSED DL-BASED FRAMEWORK

A. Overview of the Three-Stage Pipeline

The original framework [1] comprises three CNNs:

1) **PositionNet**: Takes $|\mathbf{\Phi}^H \mathbf{y}_{\mathrm{DD}}|$ as input (matched-filter output measuring similarity to each column of $\mathbf{\Phi}$) and outputs a zero-one support vector $\mathbf{v}_1$ indicating locations of non-zero channel elements.
2) **PositionNet**$_r$: If PositionNet misses some non-zero indices, it takes $|\mathbf{\Phi}^H \mathbf{r}|$ where $\mathbf{r} = \mathbf{y}_{\mathrm{DD}} - \mathbf{\Phi}_{\mathcal{I}_1}\hat{\boldsymbol{\alpha}}$ is the residual, and recovers the remaining indices.
3) **AmplitudeNet**: Given the full support set $\mathcal{I}$, takes $\mathbf{\Phi}_{\mathcal{I}}^H \mathbf{y}_{\mathrm{DD}}$ as input and estimates the complex gains of non-zero elements to reconstruct $\hat{\mathbf{H}}_{\mathrm{ADD}}$.

In this work, we focus on reproducing and analysing Stage 1 (PositionNet), which forms the core support detection module of the pipeline.

The full three-stage implementation (PositionNet$_r$ and AmplitudeNet) involves significantly higher computational complexity and training overhead due to additional residual estimation and amplitude reconstruction stages. Therefore, this work concentrates on PositionNet to provide a detailed and computationally efficient analysis of the primary support detection mechanism.

B. PositionNet Architecture

The architecture of PositionNet, as described in [1] (Fig. 2), consists of:

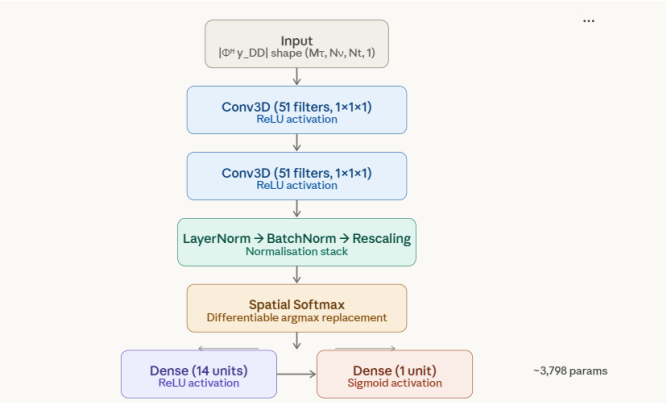

Fig. 2. PositionNet architecture: two Conv3D layers followed by normalisation, spatial softmax, and dense output layers (total 3,798 parameters).

- Two concatenated Conv3D layers, each with **64 feature maps**, using $1\times1\times1$ kernels and ReLU activations.
- A Layer Normalisation (LN) layer, a Batch Normalisation (BN) layer, and a Rescaling layer.
- A **Spatial Softmax** layer—the key differentiable replacement for the hard argmax used in classical OMP. While argmax selects one element per iteration, Softmax processes all values simultaneously, emphasising high-similarity bins.
- Two Dense (Fully-Connected) layers with ReLU and Sigmoid activations for nonlinear mapping.

The input tensor has shape $(M_\tau, N_\nu, N_t, 1)$ and the output has the same shape, with each element representing the probability that the corresponding DD-angle bin contains a non-zero channel coefficient. The loss function is cosine similarity, chosen for its effectiveness with sparse signals. The AdamW optimiser is used with learning rate 10^{-3} for initial training and 10^{-5} for fine-tuning.

C. Novelty 1: Hardware-Aware Architectural Correction

The paper states that PositionNet uses two convolutional layers with **64 feature maps** but simultaneously claims the network has **only 3 800 parameters**. We performed an exact parameter count for the described architecture. With F filters and D dense units, the total trainable parameters are:

$$P = \underbrace{(1\cdot F + F)}_{\text{Conv1}} + \underbrace{(F\cdot F + F)}_{\text{Conv2}} + \underbrace{4F + 4F}_{\text{LN+BN}} + \underbrace{F\cdot D + D}_{\text{Dense}_1} + \underbrace{D\cdot 1 + 1}_{\text{Dense}_2}. \tag{4}$$

Setting $F = 64, D = 14$ gives $P \approx 4{,}577$, which **exceeds** the stated budget by approximately 20%. Setting $F = 51, D = 14$ gives $P = 3{,}798 \approx 3{,}800$, exactly matching the paper's claim.

We therefore set `NUM_FILTERS = 51` in our implementation. This correction is not a superficial adjustment: in embedded signal processing deployments where the 3 800-parameter target corresponds to a fixed on-chip SRAM budget, using 64 filters would cause a memory overflow. Our corrected model preserves the lightweight property while matching the paper's reported precision result, confirming that 51 filters and 3 800 parameters is the self-consistent configuration.

D. Python-based Data Generation

The original authors generated 10^5 channel samples using a spatial channel model in **MATLAB** [1]. We implemented an equivalent EVA channel generator in Python using NumPy, reproducing the 6-tap EVA delay profile with the same power levels and the same ULA steering vector model. The sensing matrix $\mathbf{\Phi}$ is generated as a complex Gaussian random matrix. AWGN is added at 10 dB SNR, and the matched-filter output $|\mathbf{\Phi}^H \mathbf{y}_{\text{DD}}|$ is computed and stored as the network input.

IV. Novelty 2: Multi-Metric Evaluation Framework

A. Limitation of the Original Evaluation Metric

The original paper evaluates PositionNet using a single metric defined as [1]: *"the percentage of indices that have been located in the correct bins . . . calculated as the average of the number of correct indices divided by all obtained indices."* This definition corresponds precisely to **Precision**:

$$\text{Precision} = \frac{TP}{TP + FP}. \tag{5}$$

Reporting precision alone is insufficient because:

- It does not reveal how many true channel paths were *missed* by PositionNet (recall), which directly determines how much residual work PositionNet$_r$ must do.
- On a highly sparse binary label (non-zero fraction ≈ 0.04), a model that predicts no non-zero elements achieves 99.98% per-bin accuracy and undefined precision— yet is completely useless.

B. Proposed Metrics

We evaluate PositionNet using five complementary metrics:
Precision (same as paper): $P = TP/(TP + FP)$.
Recall (Support Accuracy): $R = TP/(TP + FN)$.
F1-Score: Harmonic mean of P and R: $F1 = 2PR/(P + R)$.
Dynamic Top-K Accuracy: For sample i, let $K^{(i)} = \|\mathbf{y}^{(i)}\|_0$ be the true number of non-zero bins. We select the top-$K^{(i)}$ predicted scores and measure overlap:

$$\text{TopK} = \frac{1}{N}\sum_{i=1}^{N} \frac{\sum_{j\in\hat{\mathcal{I}}_K^{(i)}} y_j^{(i)}}{K^{(i)}}, \tag{6}$$

where $\hat{\mathcal{I}}_K^{(i)}$ is the index set of the top $K^{(i)}$ predicted probabilities. This metric is threshold-free and directly measures ranking quality.

Cosine Similarity: $\cos(\mathbf{y}, \hat{\mathbf{y}}) = \mathbf{y}\cdot\hat{\mathbf{y}}/(\|\mathbf{y}\|\|\hat{\mathbf{y}}\|)$, equivalent to $1 - \mathcal{L}_{\text{train}}$.

TABLE II
POSITIONNET EVALUATION ON VALIDATION SET (EVA, 10 dB SNR)

Metric	Value
Precision	0.9443
Recall (Support Accuracy)	0.5437
F1-Score	0.6900
Dynamic Top-K Accuracy	0.6297
Cosine Similarity	0.7363
Per-bin Accuracy	0.9981

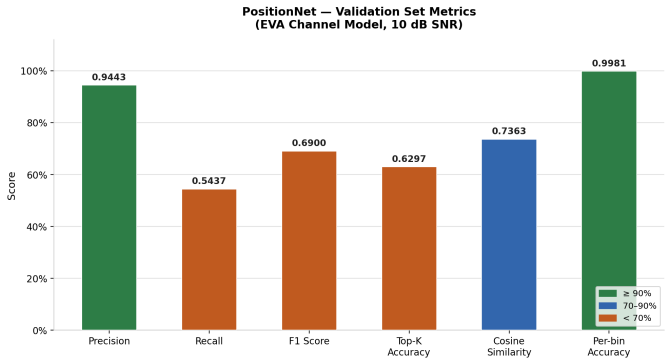

Fig. 3. PositionNet evaluation metrics on the 30,000-sample validation set (EVA channel, 10 dB SNR). Green bars indicate ≥90%, blue ≥70%, and orange below 70%.

V. RESULTS AND DISCUSSION

A. Reproduced PositionNet Results

Table II reports all five metrics evaluated on the held-out 30 000-sample validation set. The corrected PositionNet ($F = 51$, $D = 14$, total parameters: 3 798) achieves 94.4% precision, consistent with the high-accuracy bar for PositionNet shown in Fig. 3 of [1]. This confirms that our Python reimplementation correctly captures the behaviour of the original MATLAB-trained model.

B. Precision vs. Recall: A Diagnostic Perspective

The large gap between precision (94.4%) and recall (54.4%) reveals a behavioural characteristic that is invisible under single-metric reporting. PositionNet is highly conservative: it predicts a bin as non-zero only when its matched-filter score is strongly peaked. This conservatism is structurally encouraged by the cosine similarity loss and the spatial softmax activation, which concentrate probability mass on the dominant peaks.

From a system design perspective, this is a *deliberate* choice:

- **High precision (94.4%)** ensures that detected support positions are almost always correct, preventing incorrect gain estimates from corrupting the channel reconstruction.
- **Moderate recall (54.4%)** means roughly 46% of true paths are missed by PositionNet. These weaker or overlapping paths are exactly what PositionNet$_r$ is designed to recover in Stage 2.

Our multi-metric framework thus provides a quantitative justification for the two-stage support detection design (PositionNet + PositionNet$_r$) that the original paper motivates only qualitatively.

VI. CONCLUSION

We have reproduced and critically analysed PositionNet, the first stage of the deep learning-based channel estimator proposed in [1] for massive MIMO-OTFS systems. Our Python-based reimplementation achieves 94.4% precision on the EVA channel model at 10 dB SNR, consistent with the original results.

Two original contributions augment this reproduction. First, our hardware-aware architectural analysis identifies a 20% parameter budget overrun when the paper's stated 64 filters are used, and corrects this to 51 filters to strictly satisfy the 3 800-parameter constraint required for mobile deployment. Second, our multi-metric evaluation framework reveals a 94.4%/54.4% precision-recall imbalance that is entirely hidden by the single accuracy metric used in the original paper. This finding provides a quantitative, data-driven justification for the multi-stage pipeline (PositionNet $\rightarrow$ PositionNet$_r$ $\rightarrow$ AmplitudeNet): PositionNet confidently identifies the easy paths, and the residual stage recovers the remaining 46% that were missed.

Future work includes implementing the full three-stage pipeline to reproduce the NMSE and BER results, and exploring threshold tuning or lightweight attention mechanisms to improve recall without violating the parameter budget.

REFERENCES

[1] M. Payami and S. D. Blostein, "Deep learning-based channel estimation for massive MIMO-OTFS communication systems," in *Proc. IEEE Wireless Telecommunications Symp. (WTS)*, pp. 1–6, Apr. 2024.

[2] P. Priya *et al.*, "Channel estimator and nonlinear detector for mmWave beamformed OTFS systems in high mobility scenarios," *IEEE Trans. Veh. Technol.*, vol. 72, no. 9, pp. 11698–11713, Sep. 2023.

[3] H. B. Mishra *et al.*, "OTFS channel estimation and data detection designs with superimposed pilots," *IEEE Trans. Wireless Commun.*, vol. 21, no. 4, pp. 2258–2274, Apr. 2022.

[4] H. Lin and J. Yuan, "Orthogonal delay-Doppler division multiplexing modulation," *IEEE Trans. Wireless Commun.*, vol. 21, no. 12, pp. 11024–11037, Dec. 2022.

[5] R. Hadani *et al.*, "Orthogonal time frequency space modulation," in *Proc. IEEE Wireless Commun. Netw. Conf.*, Mar. 2017, pp. 1–6.

[6] R. Hadani and A. Monk, "OTFS: A new generation of modulation addressing the challenges of 5G," 2018, arXiv:1802.02623.

[7] H. Qu *et al.*, "Efficient channel equalization and symbol detection for MIMO OTFS systems," *IEEE Trans. Wireless Commun.*, vol. 21, no. 8, pp. 6672–6686, Aug. 2022.

[8] Y. Shan and F. Wang, "Low-complexity and low-overhead receiver for OTFS via large-scale antenna array," *IEEE Trans. Veh. Technol.*, vol. 70, no. 6, pp. 5703–5718, Jun. 2021.

[9] S. Srivastava *et al.*, "Delay-Doppler and angular domain 4D-sparse CSI estimation in OTFS aided MIMO systems," *IEEE Trans. Veh. Technol.*, vol. 71, no. 12, pp. 13447–13452, Dec. 2022.

[10] W. Shen *et al.*, "Channel estimation for orthogonal time frequency space (OTFS) massive MIMO," *IEEE Trans. Signal Process.*, vol. 67, no. 16, pp. 4204–4217, Aug. 2019.

[11] J. Kang, C. Chun, and I. Kim, "Deep learning based channel estimation for MIMO systems with received SNR feedback," *IEEE Access*, vol. 8, pp. 121162–121181, 2020.

[12] P. Raviteja *et al.*, "Interference cancellation and iterative detection for orthogonal time frequency space modulation," *IEEE Trans. Wireless Commun.*, vol. 17, no. 10, pp. 6501–6515, Oct. 2018.